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IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: R26-IT-011

1. Topic (12 words max)

Cross-Cultural Bias Analysis of Real-Time Emotion Detection Using Face-API.js

2. Research group the project belongs to

CEAI - Centre of Excellence in AI

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Emotion recognition using facial expressions has become an important component of modern human computer interaction systems, including mental health monitoring, e-learning platforms, adaptive user interfaces, and social robotics. Most browser-based emotion recognition solutions rely on pre-trained deep learning models such as Face-API.js. However, these models are often trained on limited and culturally homogeneous datasets, which can affect their ability to generalize across diverse populations.

Facial expressions vary significantly across cultures, influencing how emotions are expressed, perceived, and intensified. Differences in expression patterns, emotional intensity, and reaction timing can lead to misclassification or unreliable confidence scores. Despite this, many existing studies evaluate emotion recognition systems primarily based on overall accuracy, overlooking cultural variation, demographic bias, and real-time performance.

Demographic attributes such as age and gender further interact with cultural norms, potentially introducing hidden biases in pre-trained models. These biases raise fairness and ethical concerns, particularly in sensitive real-world applications. Additionally, system latency is often neglected, even though real-time emotion recognition requires both accurate and timely predictions. Delayed or unstable responses can negatively impact user trust and perceived system reliability.

This research aims to address these gaps by evaluating Face-API.js emotion recognition performance across cultural groups, emotion intensity levels, demographic attributes, and reaction-time constraints, contributing to more transparent, fair, and human-aware browser-based emotion recognition systems.

6. Brief description of Existing Research and Systems (300 words max)

Existing research in facial emotion recognition has primarily focused on developing and training deep learning models to classify human emotions using facial expressions. Many widely used systems are built upon convolutional neural networks trained on benchmark datasets such as FER-2013, CK+, and AffectNet. These models have demonstrated promising accuracy under controlled conditions and are commonly integrated into applications related to human–computer interaction, sentiment analysis, and behavioral monitoring.

Browser-based emotion recognition frameworks, including Face-API.js, have gained popularity due to their accessibility and ease of integration into web applications. Face-API.js utilizes pre-trained deep learning models to perform real-time facial expression analysis directly in the browser without requiring server-side computation. Existing systems using Face-API.js typically focus on real-time emotion detection and visualization, emphasizing usability and responsiveness rather than in-depth performance evaluation.

However, most existing research evaluates emotion recognition systems using overall accuracy or emotion classification rates, often ignoring cultural diversity, demographic variability, and fairness considerations. The majority of studies assume that facial expressions are universally consistent, which can lead to biased or misleading conclusions when systems are applied to diverse user populations. Additionally, limited attention has been given to emotion intensity analysis and probability score distributions, which provide deeper insight into model confidence and expression subtlety.

Furthermore, system-level performance factors such as detection latency, prediction stability, and reaction time are rarely analyzed, despite their importance in real-time applications. As a result, existing systems lack comprehensive benchmarking frameworks that evaluate cultural bias, demographic influence, and temporal performance simultaneously.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

The proposed solution is a browser-based real-time emotion recognition benchmarking system designed to evaluate the performance, fairness, and responsiveness of Face-API.js across diverse user groups. The system operates entirely on the client side using a standard webcam, enabling real-time facial expression analysis without server-side image storage, thereby preserving user privacy.

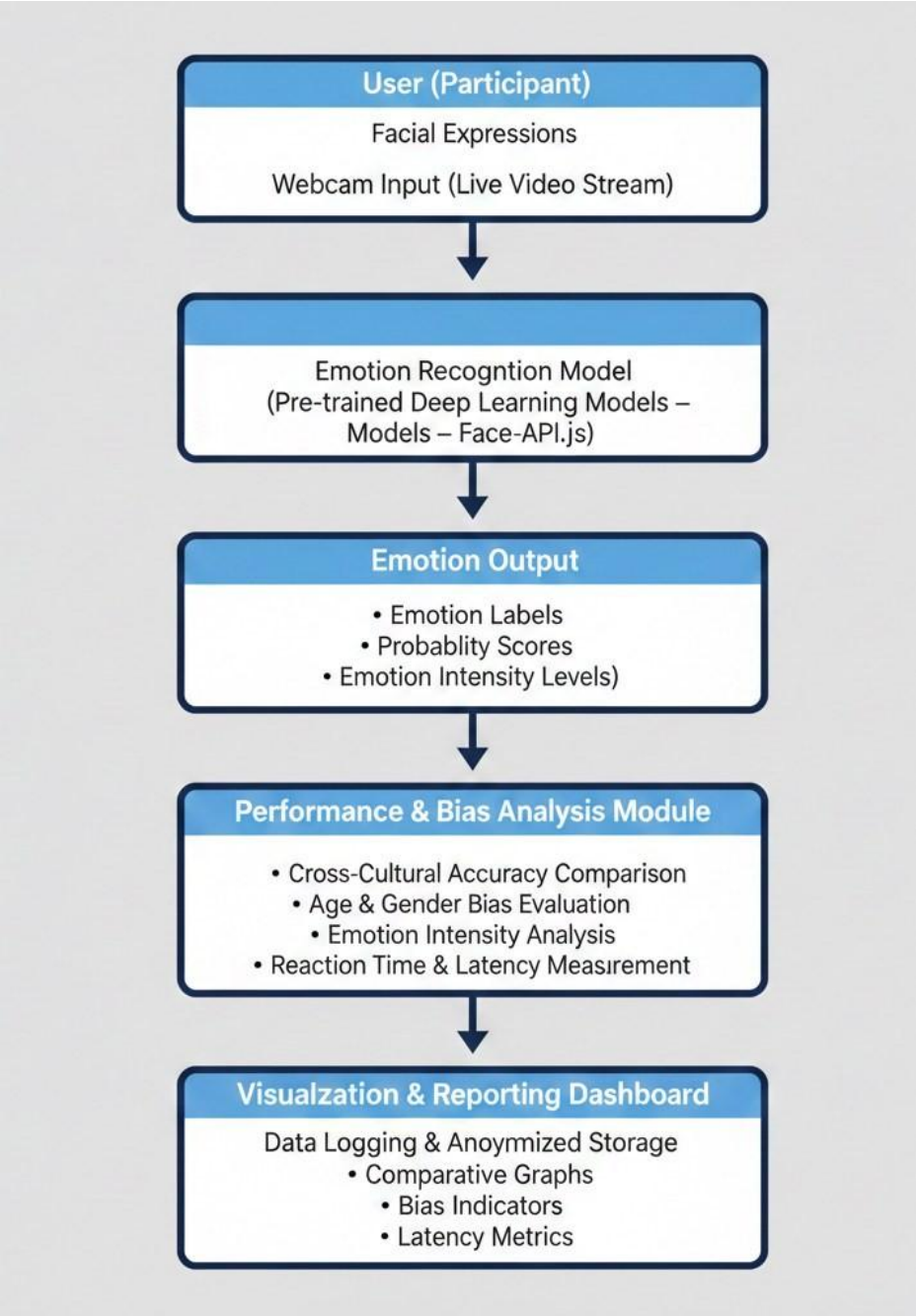
The system captures live video streams and extracts facial landmarks using Face-API.js. Based on these landmarks, pre-trained deep learning models generate emotion predictions along with corresponding probability scores. Instead of treating emotion recognition as a binary classification task, the system continuously records emotion confidence values, intensity variations, and prediction stability over time.

Participants are categorized based on cultural background, age group, and gender, allowing structured comparative analysis. For each session, the system logs detected emotions, confidence distributions, and time taken to recognize changes in facial expressions. Reaction time and latency are measured by tracking the delay between an observable facial expression change and the system's corresponding prediction update.

All collected data is anonymized and stored in structured datasets for post-processing and statistical analysis. Visualization dashboards are used to compare accuracy, bias patterns, and latency behavior across different demographic and cultural groups. This enables the identification of systematic biases, confidence inconsistencies, and performance degradation under real-world conditions.

The conceptual diagram illustrates the overall workflow of the system. It begins with live facial input via a webcam, followed by face detection and landmark extraction using Face-API.js. The extracted features are passed to the emotion recognition model, which outputs emotion probabilities and intensity scores. A performance analysis module records accuracy, bias indicators, and latency metrics. Finally, a data visualization and reporting layer presents comparative insights across cultural and demographic dimensions.

This solution does not aim to redesign emotion recognition models but rather provides a comprehensive evaluation framework that highlights their limitations. By combining real-time analysis, fairness assessment, and temporal evaluation, the proposed system contributes toward more transparent, ethical, and human-aware deployment of browser-based emotion recognition technologies.


User (Participant)
Facial Expressions
Webcam Input (Live Video Stream)
Emotion Recogntion Model
(Pre-trained Deep Learning Models –
Models – Face-API.js)
Emotion Output
• Emotion Labels
• Probablity Scores
• Emotion Intensity Levels)
Performance & Bias Analysis Module
• Cross-Cultural Accuracy Comparison
• Age & Gender Bias Evaluation
• Emotion Intensity Analysis
• Reaction Time & Latency Measurement
Visualzation & Reporting Dashboard
Data Logging & Anoymized Storage
• Comparative Graphs
• Bias Indicators
• Latency Metrics

8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

The successful development of a browser-based emotion detection system relies on access to specialized domain expertise, knowledge in multiple interdisciplinary areas, and high-quality data. Firstly, expertise in computer vision and deep learning is essential. This includes proficiency in convolutional neural networks (CNNs), transfer learning techniques, and frameworks such as TensorFlow, PyTorch, and Face-API.js. Researchers with backgrounds in these areas can effectively design models that can recognize subtle facial cues, map them to emotion categories, and optimize the system for browser-based deployment. Additionally, knowledge of front-end development and real-time web technologies is necessary to implement a responsive and interactive user interface capable of handling live video input without significant latency.

Equally important is domain knowledge in psychology and affective computing, particularly in understanding the nuances of human emotions, expression patterns, and cultural variations in emotional display. This expertise ensures that the system does not just classify facial expressions based on general patterns but incorporates context-aware interpretations that reflect cultural and individual differences. Collaborations with psychologists or behavioral scientists can enhance model accuracy and prevent misclassification, which is a known limitation in conventional emotion detection models.

Regarding data availability, extensive annotated datasets of facial expressions form the backbone of model training and evaluation. Publicly available datasets such as FER-2013, CK+, JAFFE, and AffectNet provide large-scale, labeled examples of different emotional states, often including variations in age, gender, and ethnicity. However, to address cross-cultural emotion recognition, additional data collection or augmentation may be required to capture underrepresented demographic groups and subtle expression differences. Access to these datasets, combined with techniques such as data augmentation, synthetic data generation, and domain adaptation, ensures that the system can generalize well to real-world scenarios.

Finally, collaborative knowledge-sharing platforms, open-source libraries, and prior research studies provide essential guidance for system development. These resources allow the integration of best practices in preprocessing, model optimization, and browser-based deployment while avoiding common pitfalls such as overfitting, bias, or latency issues. Moreover, ethical considerations surrounding user privacy, consent, and data security can be informed by existing frameworks in AI and human-computer interaction, ensuring the responsible use of emotion detection technologies.

In summary, the availability of specialized expertise in AI, computer vision, front-end development, and psychology, combined with extensive publicly accessible and culturally diverse datasets, provides a strong foundation for this emotion detection research. These resources collectively enable the creation of a robust, accurate, and socially aware browser-based system capable of interpreting human emotions effectively in diverse contexts

9. Objectives and Novelty

Main Objective To evaluate and benchmark the cultural, demographic, and temporal reliability of Face-API.js emotion recognition in real-time browser environments.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Dissanayake D.M.D.N IT22541970	Cross-Cultural Emotion Detection Accuracy Tasks	• Measure emotion classification accuracy across cultural groups • Compare confusion patterns between cultures	• First browser-based cultural benchmarking of Face-API.js emotion detection
H.K.G.Y.Perera IT22254252	Emotion detection and pattern analysis	• Analyze probability scores instead of binary outputs • Identify subtle expression intensity variations	• Moves beyond accuracy to fine-grained emotional confidence analysis
Deshan V.W.S IT22278876	Demographic Bias Evaluation (Age & Gender) Tasks	• Evaluate performance differences across age and gender groups • Identify intersectional bias (culture × age × gender)	• Reveals hidden fairness issues in pre-trained emotion models

Gunawardhana W.M.R.T IT22276728	Emotion Reaction Time & Latency Evaluation Tasks	• Measure detection delay when expressions change • Analyze prediction stability and lag	• Introduces temporal accuracy as a core evaluation metric

10. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	

b) The proposed topic exhibits novelty.

Yes	✓	No	

c) The student group can successfully execute the proposed project.

Yes	✓	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	✓	No	

e) Any other comments:

Recommend proceeding with this project.

	Title	First Name	Last Name	Signature
Supervisor	Dr.	Junius Anjana	Vidanaralage	Digital Signer:Dr. Junius Anjana Date:13-Jan-26 10:03:28 +05:30
Co-Supervisor	Ms.	Pubudika	Wijesundara	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				